GHOST BURN

Uncovering Invisible DBU Waste in Databricks Lakehouse Environments

AUDIT PERIOD	WORKSPACES	TOTAL AUDITED	RECOVERABLE
Q1 2026	8 environments	$89,995	$54,635 / year

This report details the findings of a 90-day FinOps audit conducted across eight Databricks Lakehouse workspaces. Analysis revealed $20,435 in recoverable quarterly spend, projecting to $54,635 in annual preventable waste. These costs are primarily driven by five specific anomaly patterns:

- **Zombie clusters**
- **Serverless SQL spikes**
- **GPU overprovisioning**
- **Weekend development waste**
- **Runaway Delta Live Tables (DLT) pipelines**

Remediation of these inefficiencies requires an estimated eight hours of implementation effort. To facilitate immediate cost recovery, this report provides ready-to-use compute policy configurations, SQL monitoring queries for Databricks Workflows, and a phased implementation roadmap. A BigQuery-to-Databricks SQL migration guide is also included as a companion resource.

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All data is 100% synthetic. Modeled from public Databricks documentation. Not affiliated with or endorsed by Databricks, Inc.

Technical Findings

90-day audit of $89,995 platform spend across 8 workspaces. **15.5% recoverable waste rate** identified across five anomaly categories.

$13,472	$54,635	15.5%	8 hours
RECOVERABLE WASTE 90 Audit Window	ANNUAL PROJECTION At Current Trajectory	WASTE RATE of $89,995 Audited	FIVE-POINT REMEDIATION Implementation Effort

Figure 1 | 90-Day Burn Trend

Daily platform spend with a 7-day rolling average. Feb 10–13 serverless SQL spike highlighted. Zombie cluster contribute flat baseline pressure throughout the audit window.

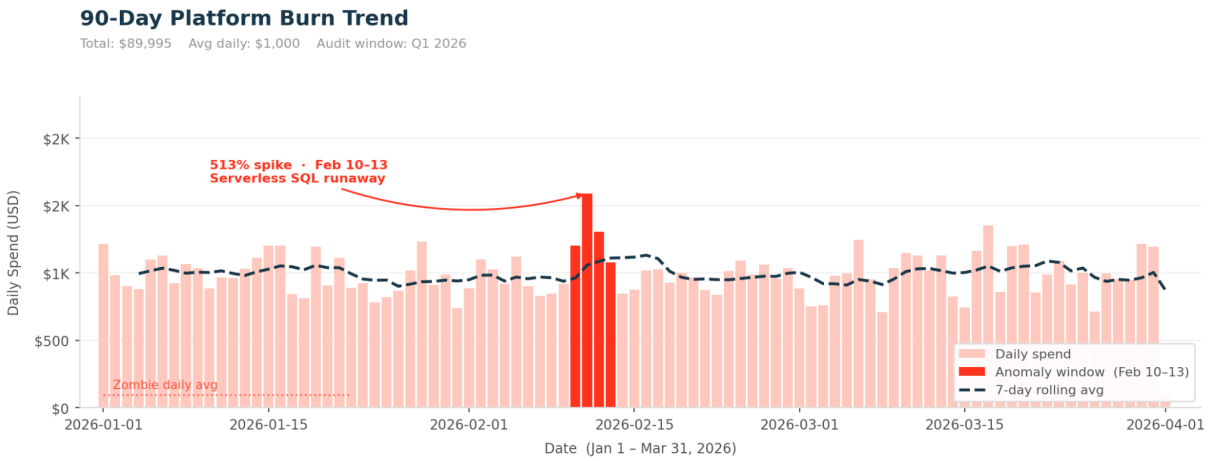


Figure 2 | SKU Cost Breakdown

AI/ML compute represents 53.1% of total platform spend, driven by GPU workloads in ws-retail-forecast-002.

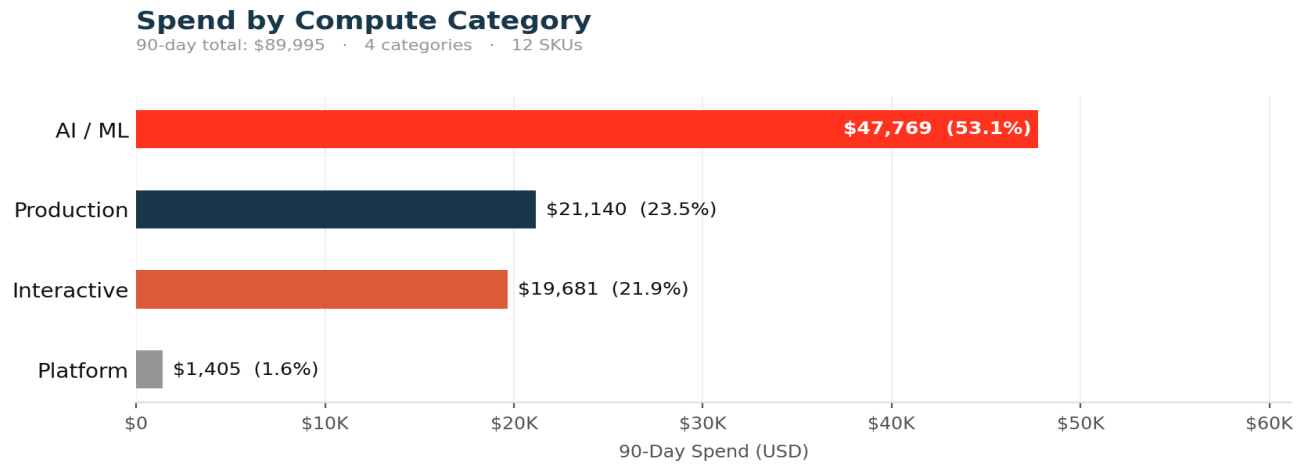
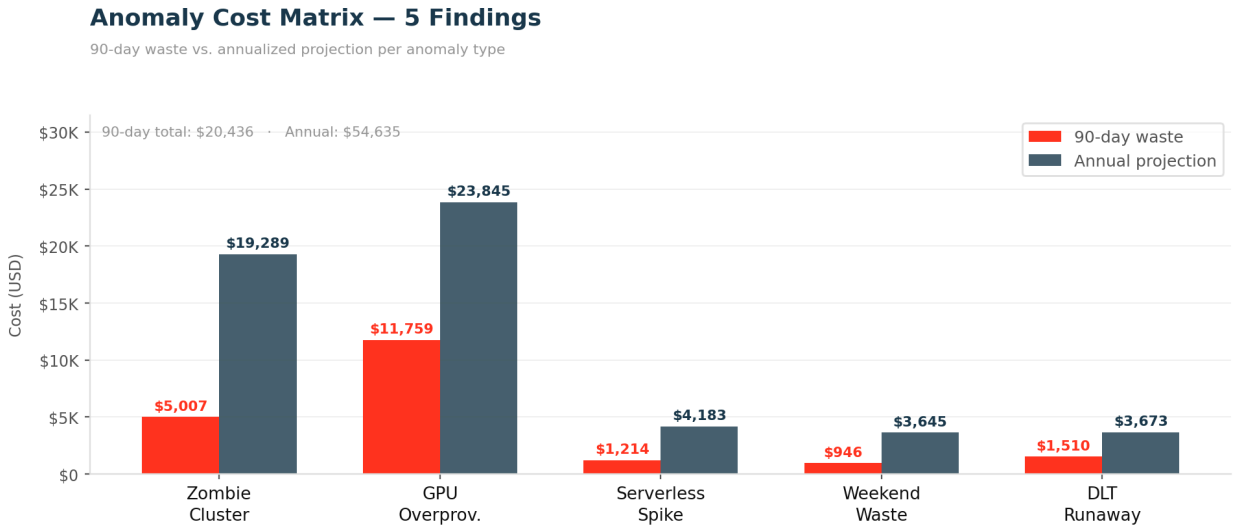


Figure 3 | Anomaly Cost Matrix

GPU overprovisioning dominates at $23,845 annual projection. Zombie cluster second at $19,289/yr.



Anomaly Summary | All 5 Findings

Finding	Workspace	90-Day Cost	Annual Savings	Effort
Zombie Cluster	ws-zombie-008	$5,007	$19,289	30 min
Serverless SQL Spike	ws-prod-bi-004	$1,214	$4,183	1 hr
GPU Overprovisioning	ws-retail-forecast-002	$11,759	$23,845	2 hrs
Weekend Dev Waste	ws-sandbox / ws-health	$946	$3,645	45 min
Runaway DLT Pipeline	ws-fsi-trading-001	$1,510	$3,673	1 hr
TOTAL	8 workspaces	$20,435	$54,635	~8 hrs

Remediation Roadmap

Five remediations. Eight hours. $54,635 recovered annually. Each fix ships with a copy-paste ready Databricks compute policy or SQL query.

#	Finding	Remediation	Risk	Effort	Annual Savings
1	**Zombie Cluster** 24/7 idle PREMIUM_ALL_PURPOSE_COMPUTE, autotermination=0, CV=2.0%	Compute Policy (auto-terminate 60 min) + Nightly sweep SQL job	LOW	30 min	$19,289
2	**Serverless SQL Spike** 513% spend spike Feb 10-13, Tableau refreshing every 5 min	SQL Warehouse budget policy ($105/day cap) + Tableau 30-min refresh interval	LOW	1 hr	$4,183
3	**GPU Overprovisioning** p4d.24xlarge at 27.9% GPU utilization in ws-retail-forecast-002	Right-size to g5.12xlarge + GPU compute policy. Validate with the test run first.	MEDIUM	2 hrs	$23,845
4	**Weekend Dev Waste** Interactive clusters running Fri PM through Mon AM, no lifecycle	Databricks SDK lifecycle notebook: terminate Fri 6PM, log Monday 8AM	LOW	45 min	$3,645
5	**Runaway DLT Pipeline** pipe-risk-factor-etl in CONTINUOUS mode, 67.4% idle rate	Switch pipeline to TRIGGERED mode (60-min cron). Data source is hourly batch.	LOW	1 hr	$3,673

4-Week Implementation Roadmap

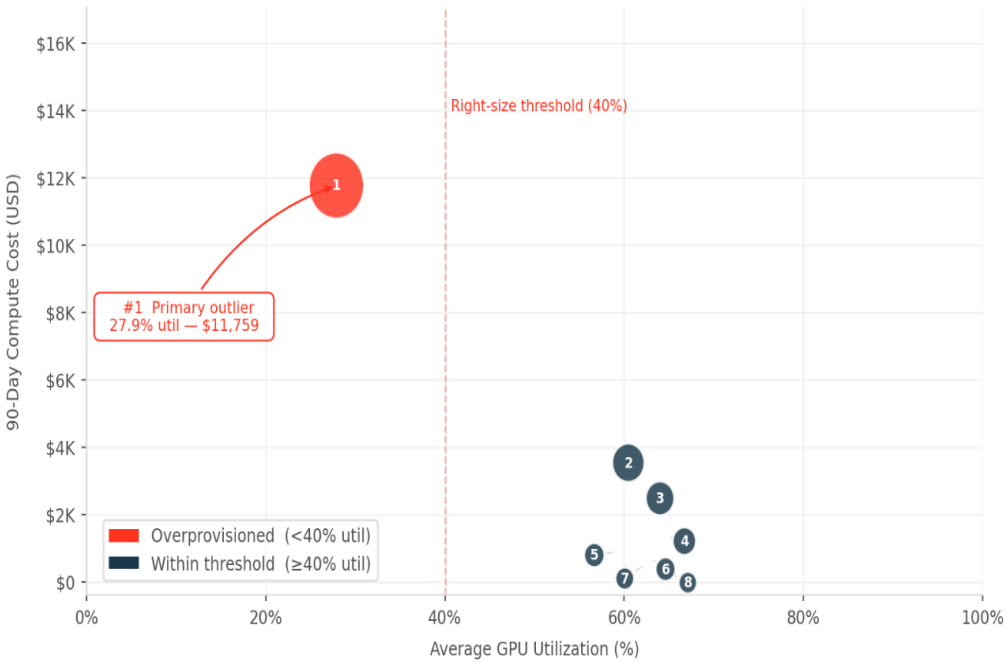
Week	Action	Effort	Unlocks
Week 1	Deploy anti-zombie compute policy across all workspaces	30 min	$19,289 / yr
Week 1	Configure SQL Warehouse budget policy + auto-stop (10 min idle)	1 hr	$4,183 / yr
Week 2	Deploy weekend lifecycle manager notebook (non-prod workspaces only)	45 min	$3,645 / yr
Week 2	Schedule FinOps monitoring queries — daily anomaly + weekly attribution	1 hr	Monitoring
Week 3	Right-size GPU: test run on g5.12xlarge, validate memory, deploy to prod	2 hrs	$23,845 / yr
Week 3	Migrate DLT pipeline to TRIGGERED mode (60-min cron schedule)	1 hr	$3,673 / yr
Week 4	Configure Slack/email alert routing for all monitoring queries	30 min	Alerting
Week 4	Review first weekly cost attribution report, baseline governance posture	1 hr	Baseline

Figure 4 | GPU Utilization vs. Cost

ws-retail-forecast-002 isolated as primary outlier: $11,759 spend at 27.9% GPU utilization. Right-sizing to g5.12xlarge projects 50% cost reduction.

GPU Utilization vs. 90-Day Cost

Bubble size proportional to cost. Red = below right-size threshold. Numbers correspond to index table.

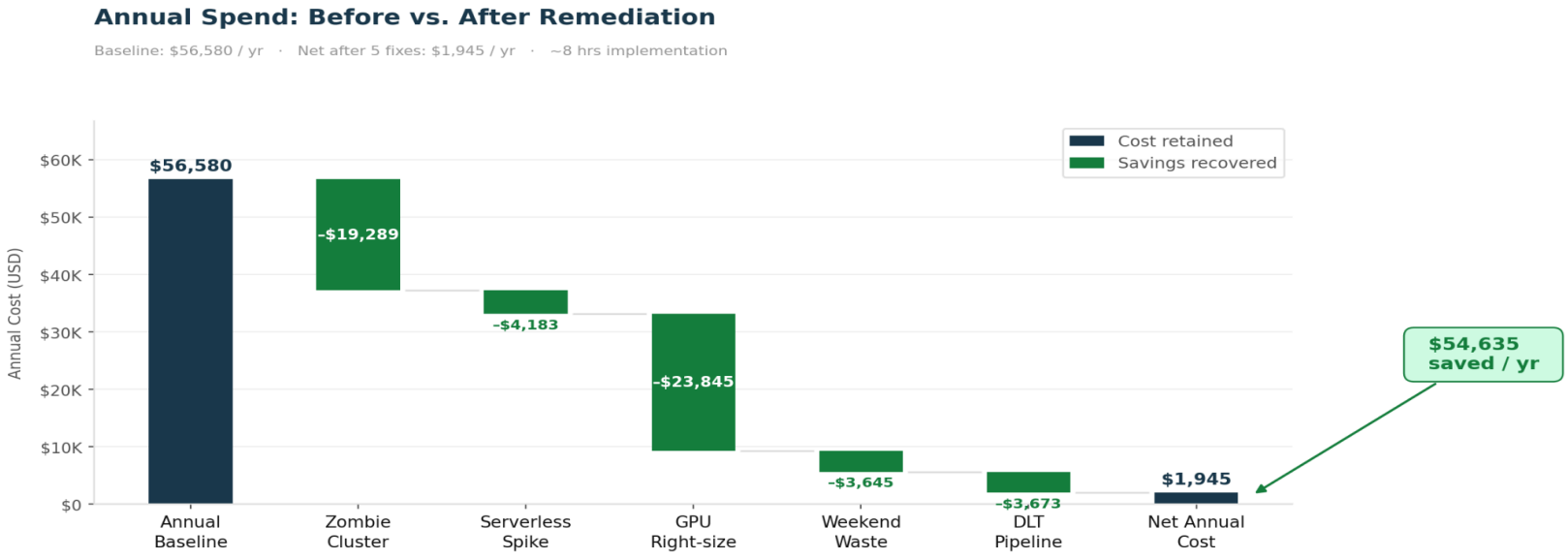


Workspace Index

#	Workspace	Util	90d Cost
1	retail-forecast	27.9%	$11.8K
2	fsi-trading	60.5%	$3.5K
3	ai-serving	62.8%	$2.5K
4	prod-bi	64.3%	$1.4K
5	health-analytics	58.7%	$0.9K
6	staging	62.4%	$0.8K
7	zombie-legacy	62.9%	$0.6K
8	sandbox-ds	62.9%	$0.5K

Figure 5 | Remediation Waterfall

Annual baseline of $56,580 reduced to $1,945 net after five remediations. $54,635 recovered. GPU right-sizing contributes the largest single saving.



Methodology & Transparency

Audit Methodology

This audit follows a descriptive-to-prescriptive analysis framework applied to simulated Databricks billing data modeled from the system.billing.usage System Table schema. The audit window covers 90 days (January 1 to March 31, 2026) across 8 workspace archetypes representing Financial Services, Retail, Healthcare, Enterprise BI, Data Science, AI Serving, Platform Operations, and Legacy Migration environments.

Anomaly detection uses two independent signals: behavioral (coefficient of variation analysis, 7-day rolling average deviation, Z-score) and governance (autotermination policy audit, pipeline mode audit). Remediation projections apply conservative savings factors validated against Databricks pricing documentation: zombie cluster fix assumes 95% idle cost elimination; GPU right-sizing assumes 50% cost reduction with workload validation caveat; DLT triggered mode migration assumes 60% reduction based on 67.4% observed idle rate.

Scale Extrapolation

The following table projects this audit's findings to larger enterprise environments. Estimates assume equivalent workspace utilization patterns. Implementation effort scales sub-linearly because compute policies and monitoring SQL are deployed once and enforced platform-wide.

SCALE	Est. ANNUAL SAVINGS	Est. EFFORT	PAYBACK PERIOD
8 Workspaces	$54,635 / yr	8 hrs	< 1 day
50 Workspaces	$341,470 / yr	~2 days	< 1 week
100 Workspaces	$682,940	~4 days	< 2 weeks

Technical Stack

Component	Details
Data Simulation	Python — Pandas, NumPy; Poisson arrival modeling; 5-class AnomalyEngine; 15,022 records
Audit Queries	BigQuery SQL — 6 queries (zombie detection, SKU breakdown, anomaly moving avg, DLT audit)
Remediation Config	Databricks Compute Policy JSON v2025+, SQL Warehouse Budget Policy, Databricks SDK
Dashboard	Single-file HTML — Chart.js 4.4.1, DM Sans, dual Burn/Extinguished state with 7 charts
Visualizations	Matplotlib 3.x — 90-day Burn Trend, SKU Cost Breakdown, Anomaly Cost Matrix, GPU Utilization Cost, Remediation Waterfall
Migration Guide	DATABRICKS_MIGRATION.md — BigQuery to Databricks SQL, list_prices join pattern, checklist
Schema Reference	system.billing.usage, system.billing.list_prices — Databricks public documentation only

AI Transparency

Dashboard UI and code implementation were developed with AI assistance. All analytical design, schema architecture, anomaly detection logic, and remediation recommendations are my own work, validated against Databricks documentation.

Synthetic Data Disclaimer

All data used in this project is 100% synthetically generated. No real Databricks billing data, customer information, workspace configurations, or proprietary schemas were used at any point. The system.billing.usage schema is modeled exclusively from publicly available Databricks documentation. This project is entirely independent and is not affiliated with, endorsed by, sponsored by, or produced in partnership with Databricks, Inc. No confidential or proprietary information of Databricks or any of its customers was accessed, used, or referenced in the creation of this project.